Car Insurance Claim(DVT)

A Project Report

Submitted to

greatlearning
Learning for Life

By

Chaitra M

in Partial Fulfillment of
PGPDS.O.Apr25.A

TEXAS McCombs
The University of Texas at Austin
McCombs School of Business

Table Of Content

1. **Business Context:**
2. **Objective:**
3. **Data Dictionary**
4. **Creation of New Calculated Filed**
   Age Group Computation
   Claim Risk Level Computation
   Vehicle Age Group Computation
   Driver Experience Level Computation
   Claim Frequency Computation
   Claim Severity Index Computation
   Income Range Computation
   Prior Claim Computation
5. **Parameter Creation**
   Creation Of Education Level parameter
   Creation Of Education Level Filter
   Creation of Top N Parameter
   Creation of Value To Show Computation
   Creation of Dynamic Measure Field
6. **Creation Of Influence Of Driver's Education On Other Factors DashBoard**
7. **Creation Of Driver Demographics & Claims Analysis DashBoard**
8. **Creation of Driver Demographics, Education Factor & Insurance Claim Insights StoryBoard**
   1.1 Claim by Education
   1.2 Occupation Vs Education
   1.3 Gender Vs Age Group
   1.4 Income Range Vs Car Type
   1.5 Education Vs Income Range
   1.6. Claim Amount Vs Driver Experience Level
   Summary & Recommendations

Overall Insights & Recommendations
Story Conclusion
9. **Creation of Influence of Driver's Income On Other Factors Dashboard**
10. **Creation of Car Insurance Claim Analysis DashBoard**
11. **Creation Of Car Insurance Claims & Driver's Income StoryBoard**
2.1 Education Level Distribution
2.2 Customer Distribution by Car Age
2.3 Customer Income & Gender Distribution Across Birth Years
2.4 Claim Frequency by Marital Status
2.5 Claim Amount by Car Type
2.6 Relationship Between Income and Home Value
2.7 Customer Distribution by Income Range
Summary & Recommendations
Story Conclusion
12. **Exploratory Data Analysis**
13. **Tableau Public URL**

Problem Statement

Business Context:

Accidents occur worldwide due to various factors such as negligent driving, adverse climatic conditions, and infrastructure issues. These incidents lead to a significant number of insurance claims, posing a challenge for insurance companies. Accurately estimating the number of accidents and subsequent claims is crucial for the sustainability and profitability of the business. By analyzing the claims data, insurance companies can anticipate future claims, optimize their financial reserves, and develop innovative policies.

Understanding the patterns and trends in claims data is equally essential. Insights derived from such analysis enable companies to offer tailored insurance policies that provide better benefits to users while optimizing premium rates. This balance ensures customer satisfaction and maximizes the company's profitability. Additionally, analyzing claims data can help in identifying fraud, improving customer segmentation, and creating dynamic pricing models based on risk profiles.

Objective:

We are tasked with leveraging data to gain actionable insights from the Car Insurance Claims dataset. Our primary objective is to explore the dataset, identify patterns, and present our findings through an interactive and visually appealing storyboard.

Data Dictionary

The dataset contains client information related to car insurance claims. It includes personal details (such as client ID, age, and gender, kids, homekids, year of job, income, education, parent details, marital status, occupation), vehicle information (car type,travel time, and usage type, car age), and claim details (old claim, claim amount, and claim frequency).

Assumption	**Car Owner and Driver are same Amounts are in Dollars ($)**
ID	Identification Variable
KIDSDRIV	Number of teenagers among the car owner's children who can drive a car.

BIRTH	Date of birth of the driver.
HOMEKIDS	No of children the car owner has.
YOJ	Years on Job.
INCOME	Income of the driver
PARENT1	Is the car owner a Single Parent
HOME_VAL	Value of the house owned by the car owner
MSTATUS	Marital status of the car owner
GENDER	Gender of the driver
EDUCATION	Maximum Education level of the driver
OCCUPATION	Occupation of the driver
TRAVTIME	Time taken to get to work on an average
CAR_USE	Purpose of using the car

BLUEBOOK	Value of the Vehicle(in dollars)
CAR_TYPE	Car type
OLDCLAIM	Total claim (in past 5 years - in dollars)
CLM_FREQ	Number of claims (in past 5 years)
CLM_AMT	If car was in a crash, what is the currently claimed amount(in dollars)
CAR_AGE	Age of car
URBANICITY	Where the car is being driven primarily

Tableau Public – Book2

File Data Window Help

Tableau Desktop Public Edition — Buy Tableau

Connections — Add

Car claims for insurance
Microsoft Excel

Sheets

Use Data Interpreter

Data Interpreter might be able to clean your Microsoft Excel workbook.

Data

Dictionary

New Union

New Table Extension

Go to Worksheet

Data Source — Sheet 1

Dictionary (Car claims for insurance)

Filters
0 | Add

Dictionary — 2 fields 21 rows — 21 rows

Name
Dictionary

Fields

Type	Field Name	Phys...	Rem...
Abc	Assumption	Dictio...	Assu...
Abc	Car Owner and Driver are same	Dictio...	Car O...

Abc Dictionary Assumption	Abc Dictionary Car Owner and Driver are :
ID	Identification Variable
KIDSDRIV	Number of teenagers among ...
BIRTH	Date of birth of the driver
HOMEKIDS	No of childern the car owner ...
YOJ	Years on Job. How many year...
INCOME	Income of the driver
PARENT1	Is the car owner a Single Par...
HOME_VAL	Value of the house owned by ...
MSTATUS	Marital status of the car owner
GENDER	Gender of the driver
EDUCATION	Maximum Education level of ...
OCCUPATION	Occupation of the driver
TRAVTIME	Time taken to get to work on ...
CAR_USE	Purpose of using the car
BLUEBOOK	What is the worth of the car. ...
CAR_TYPE	Car type

Tableau Public – Book2

File Data Window Help

Tableau Desktop Public Edition — Buy Tableau

Connections — Add

Car claims for insurance
Microsoft Excel

Sheets

Use Data Interpreter

Data Interpreter might be able to clean your Microsoft Excel workbook.

Data

Dictionary

New Union

New Table Extension

Go to Worksheet

Data Source — Sheet 1

Data+ (Car claims for insurance)

1 Alert

Filters
0 | Add

Data — 21 fields 7647 rows — 100 rows

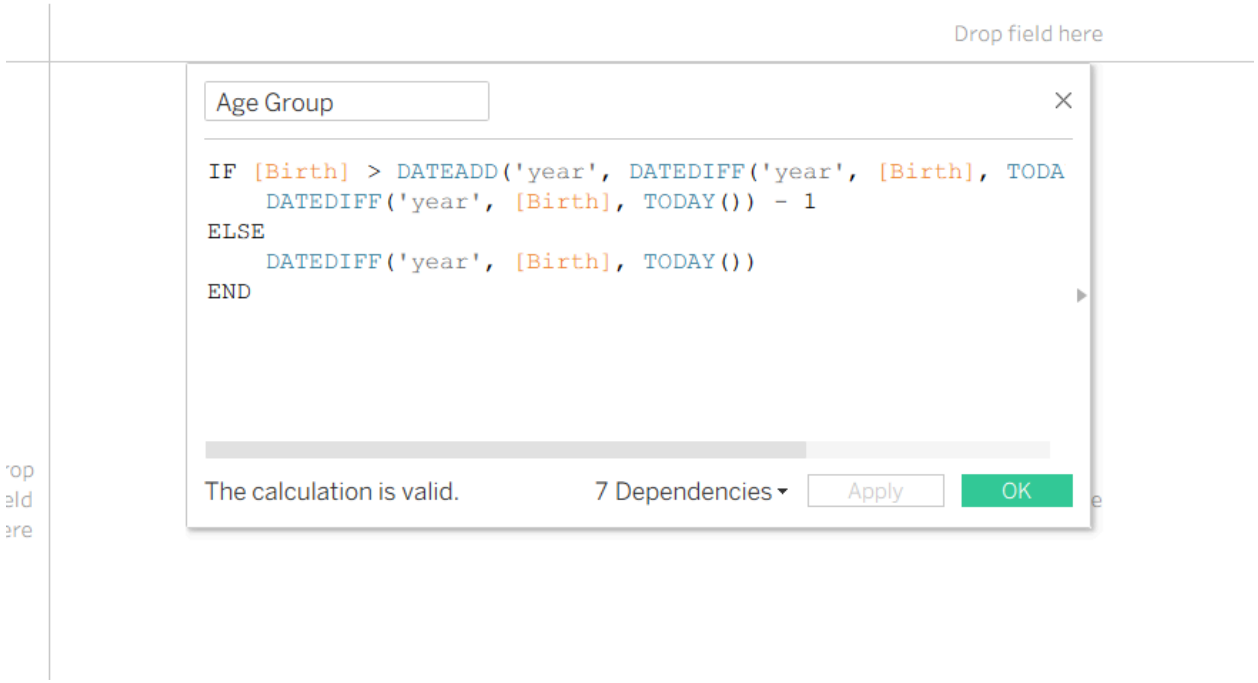
Table Details

Data ID	Data Kidsdriv	Data Birth	Data Homekids	Data YOJ	Data Income	Data Parent1	Data Home Val	Data Mstatus	Data Gender	Data Education	Data Occupation	Data Trav
100130023	0	05-02-1955	2	13	26,763	Yes	141,019	No	F	Masters	Manager	
100263241	0	05-10-1961	0	9	156,060	No	381,438	Yes	M	Bachelors	Blue Collar	
100321982	0	12-11-1959	1	11	1,965	Yes	0	No	F	PhD	Home Maker	
100391818	0	11-10-1961	1	10	62,361	No	185,738	Yes	F	Bachelors	Home Maker	
100549277	0	06-10-1966	1	10	34,192	Yes	0	No	F	High School	Blue Collar	
100550672	0	15-07-1958	2	7	17,755	Yes	148,815	No	F	<High School	Clerical	
100560602	0	21-10-1965	4	16	72,641	No	230,443	Yes	M	Bachelors	Clerical	
100698866	0	10-01-1939	0	9	100,207	No	320,199	No	M	Bachelors	Manager	
101131398	1	20-07-1936	1	2	38,111	No	152,804	Yes	F	Masters	Lawyer	
101278471	0	22-12-1948	0	0	0	No	0	No	F	<High School	Home Maker	
101504483	0	03-08-1	3	13	39,923	No	133,822	Yes	M	<High School	Blue Collar	
101597061	0	18-10-1957	0	8	94,591	No	309,644	No	M	<High School	Blue Collar	
101619581	0	13-02-1950	0	0	0	No	0	No	M	<High School	Student	
102128945	0	13-01-1955	3	17	49,098	No	157,404	Yes	M	Bachelors	Blue Collar	
102133550	0	11-04-1962	0	11	33,216	No	0	No	M	High School	Clerical	
102270088	0	08-11-1959	1	10	39,710	No	172,759	Yes	M	High School	Blue Collar	

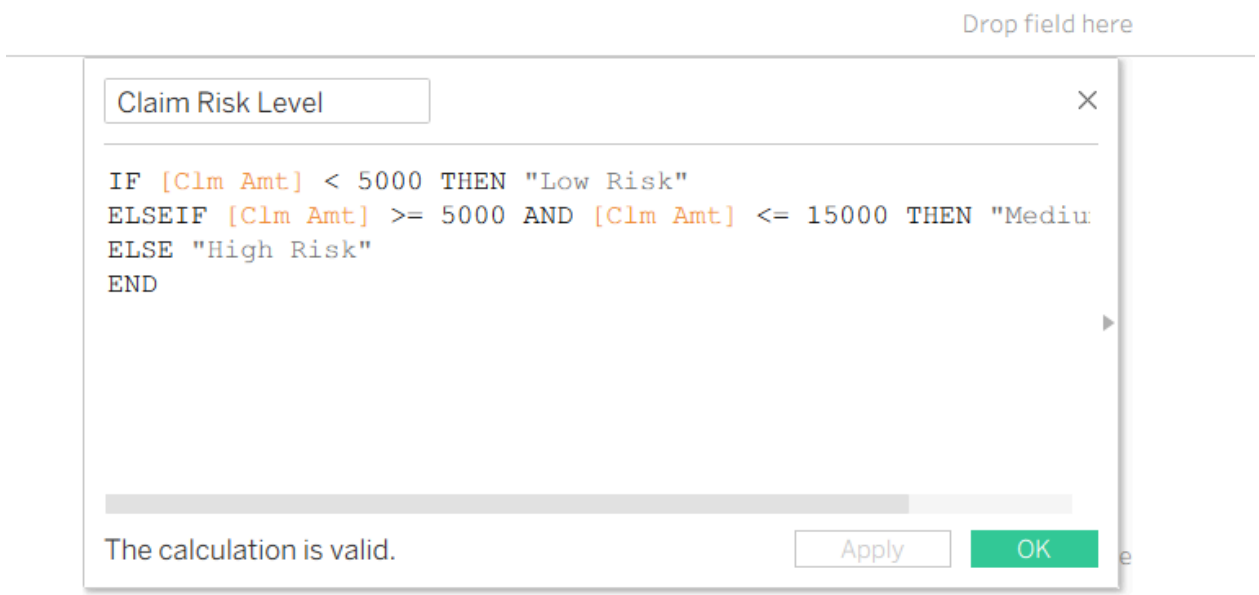
22-12-1948

Creation of New Calculated Filed

Age Group Computation



Claim Risk Level Computation



Vehicle Age Group Computation

Drop field here

Vehicle Age Group

```
IF [Car Age] < 3 THEN "New Vehicle"
ELSEIF [Car Age] >= 3 AND [Car Age] <= 7 THEN "Mid-Age Vehi
ELSE "Old Vehicle"
END
```

The calculation is valid. Apply OK

Driver Experience Level Computation

Drop field here

Driver Experience Level

```
IF [YOJ] < 2 THEN "Beginner"
ELSEIF [YOJ] >= 2 AND [YOJ] <= 5 THEN "Intermediate"
ELSE "Experienced"
END
```

The calculation is valid. 3 Dependencies Apply OK

Claim Frequency Computation

Drop field here

Claimed

```
IF [Clm Freq] > 0 THEN 1 ELSE 0 END
```

The calculation is valid. 3 Dependencies Apply OK

Claim Severity Index Computation

Drop field here

Claim Severity Index

```
IF [Bluebook] = 0 OR ISNULL([Bluebook]) THEN 0
ELSE [Clm Amt] / ([Bluebook] + 1)
END
```

The calculation is valid. 4 Dependencies Apply OK

Income Range Computation

Drop field here

IncomeRange

```
IF [Income] < 50000 THEN "<50000"
ELSEIF [Income] < 100000 THEN "50000-100000"
ELSEIF [Income] < 150000 THEN "100000-150000"
ELSEIF [Income] < 200000 THEN "150000-200000"
ELSEIF [Income] < 500000 THEN "200000-500000"
ELSEIF [Income] < 800000 THEN "500000-800000"
ELSE ">800000"
END
```

The calculation is valid. 7 Dependencies Apply OK

Prior Claim Computation

Drop field here

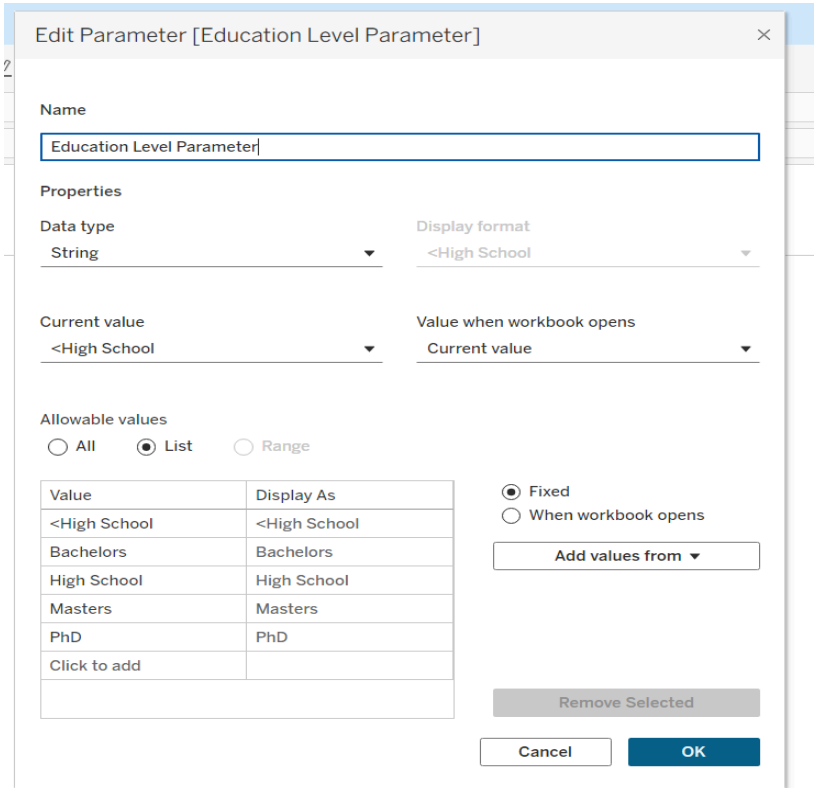
Has Prior Claim

```
IF [Oldclaim] > 0 THEN "Yes" ELSE "No" END
```

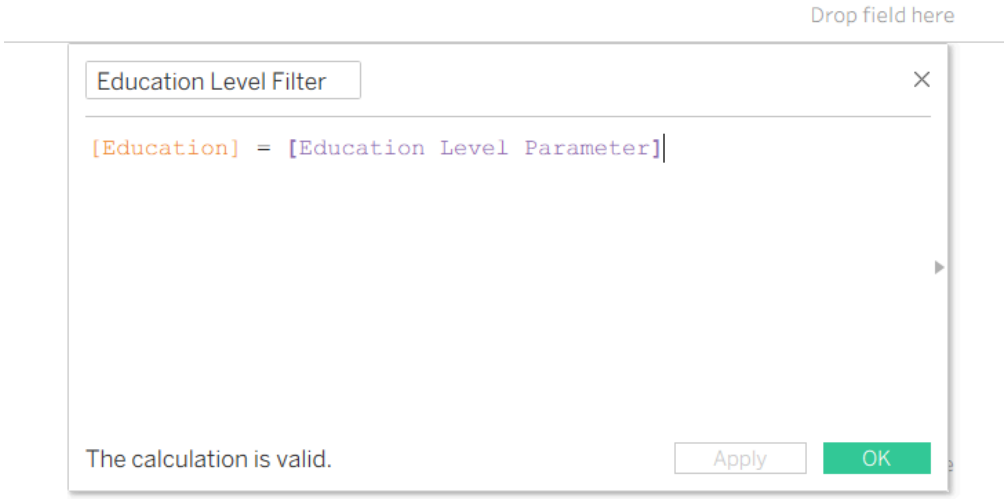
The calculation is valid. Apply OK

Parameter Creation

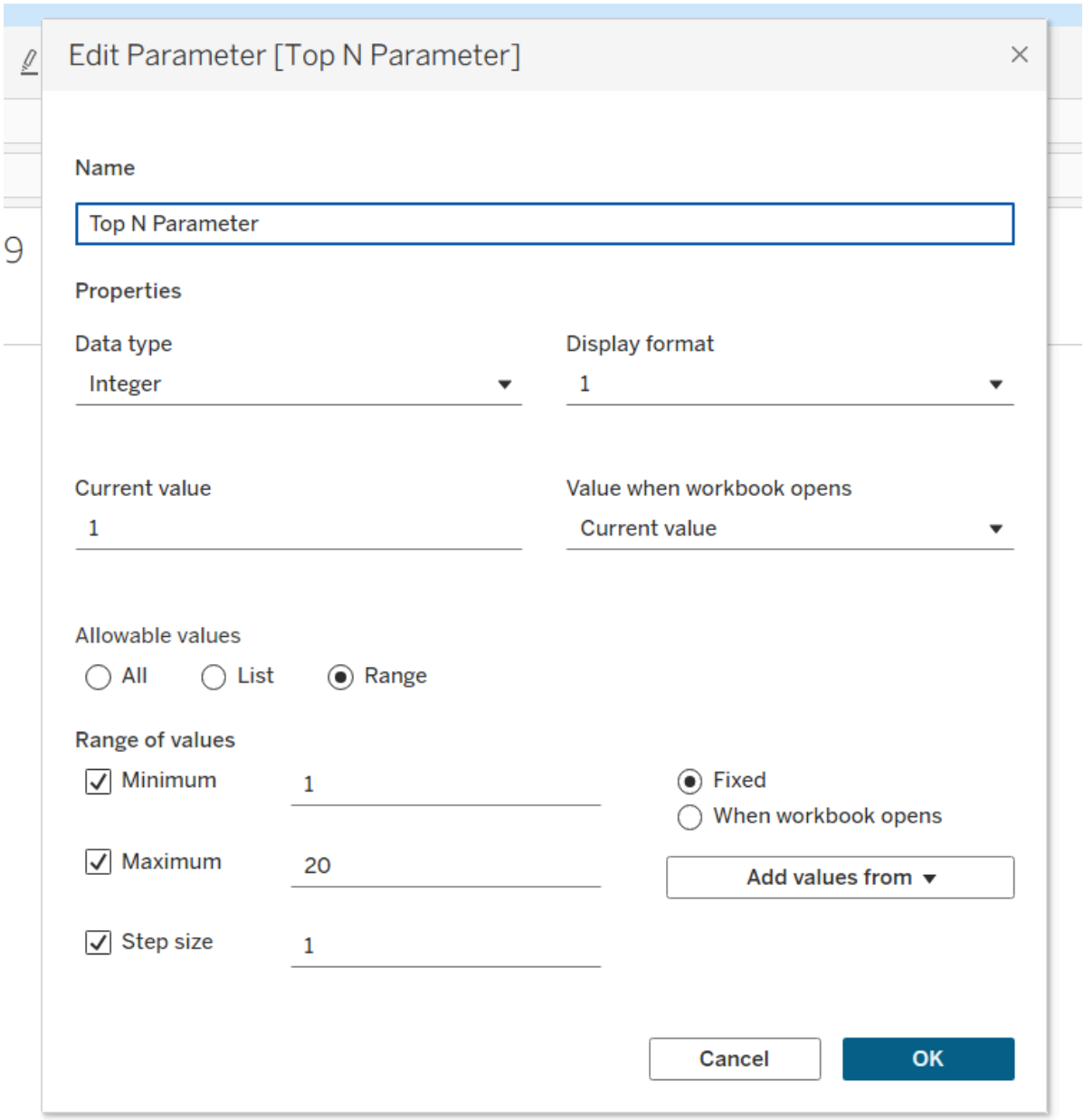
Creation Of Education Level parameter



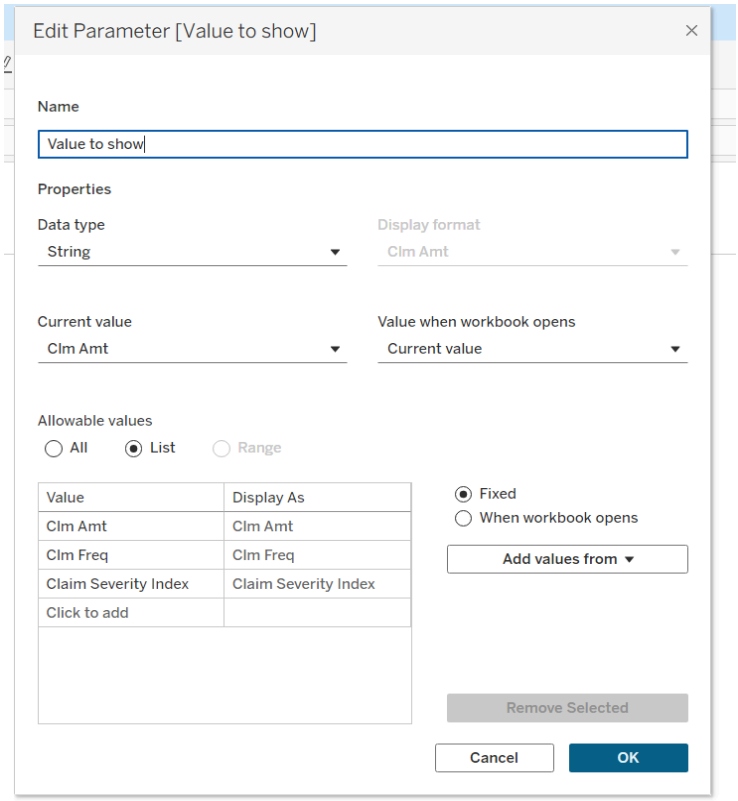
Creation Of Education Level Filter



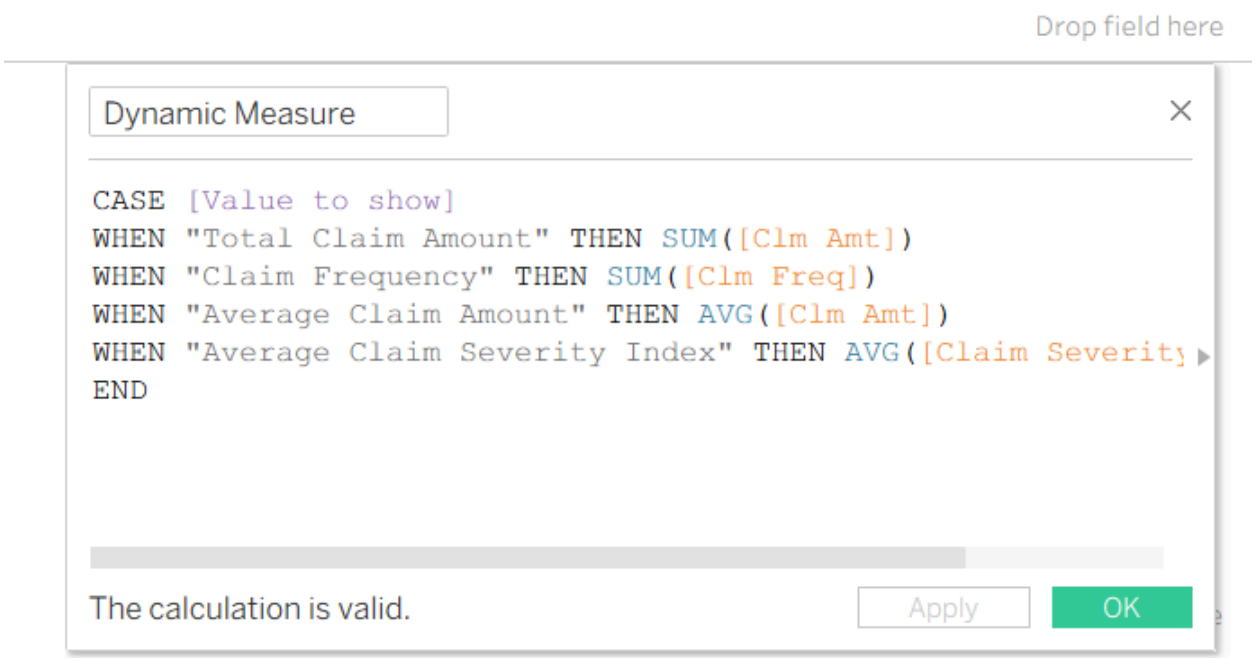
Creation of Top N Parameter



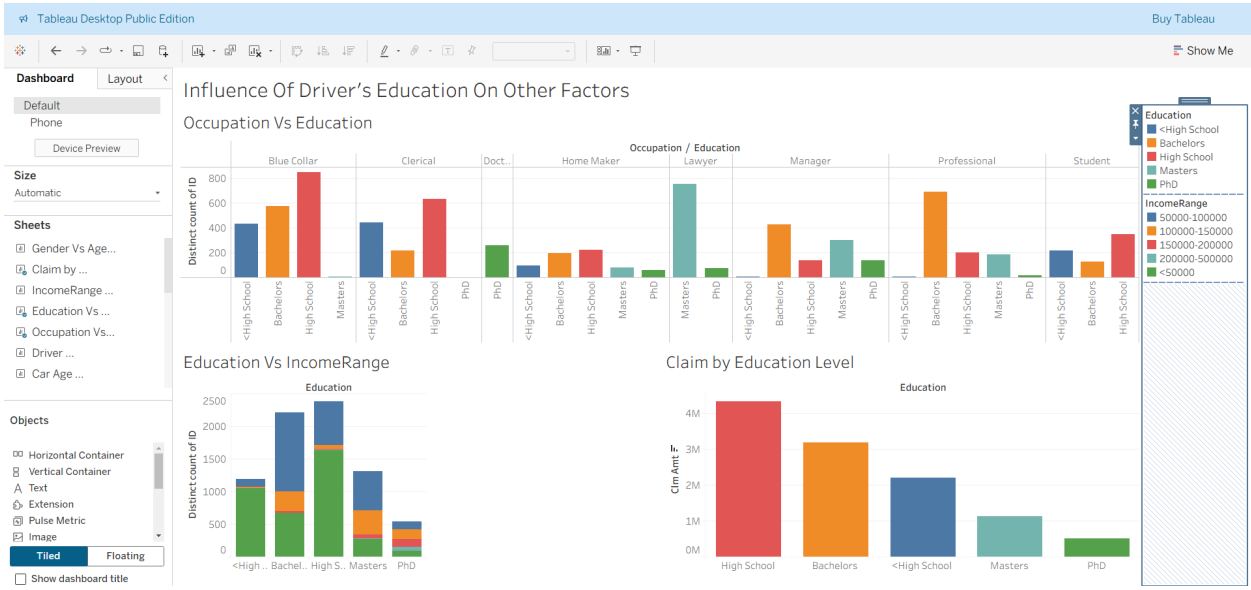
Creation of Value To Show Computation



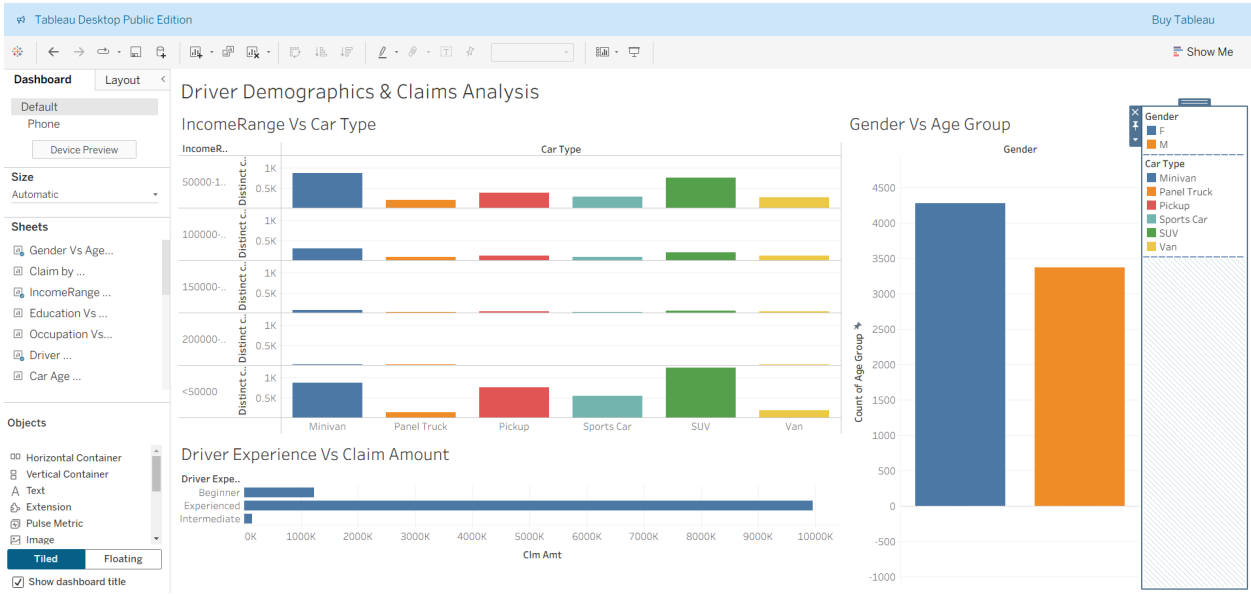
Creation of Dynamic Measure Field



Creation Of Influence Of Driver's Education On Other Factors DashBoard

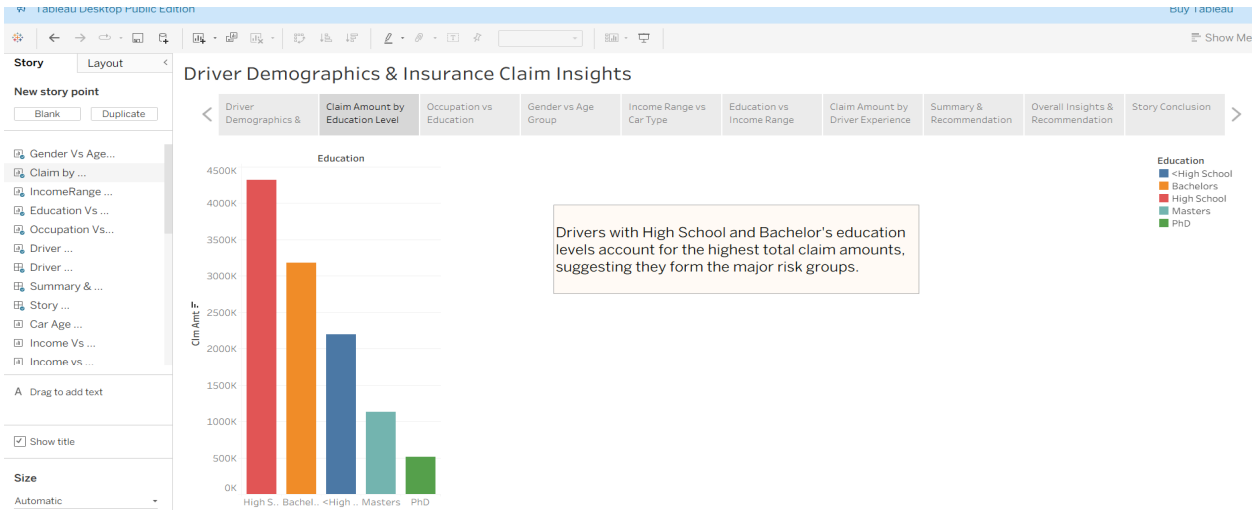


Creation Of Driver Demographics & Claims Analysis DashBoard

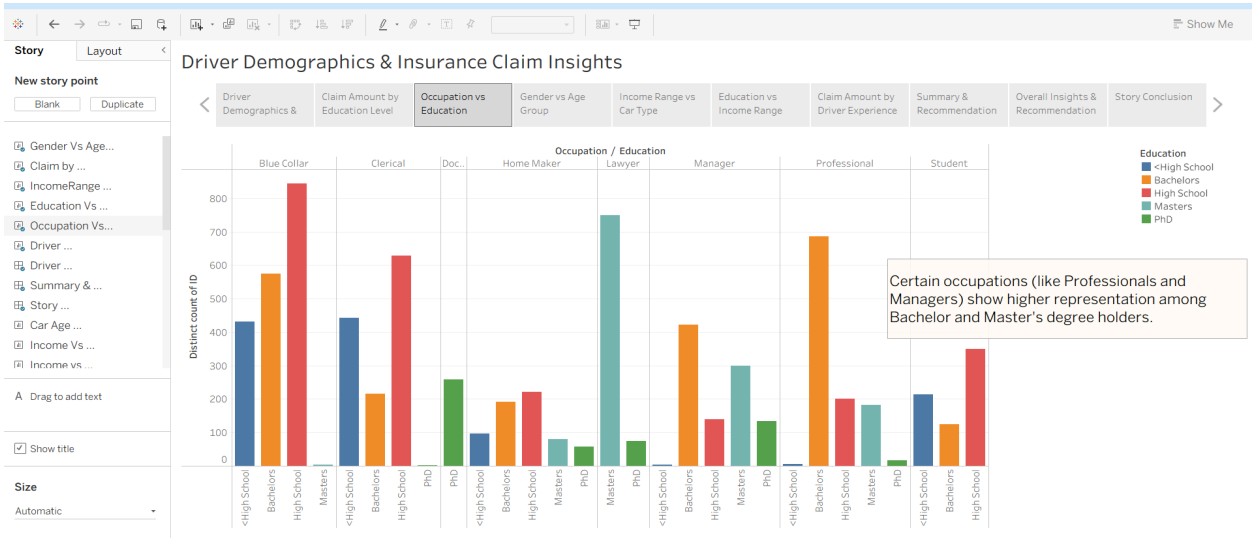


Creation of Driver Demographics, Education Factor & Insurance Claim Insights StoryBoard

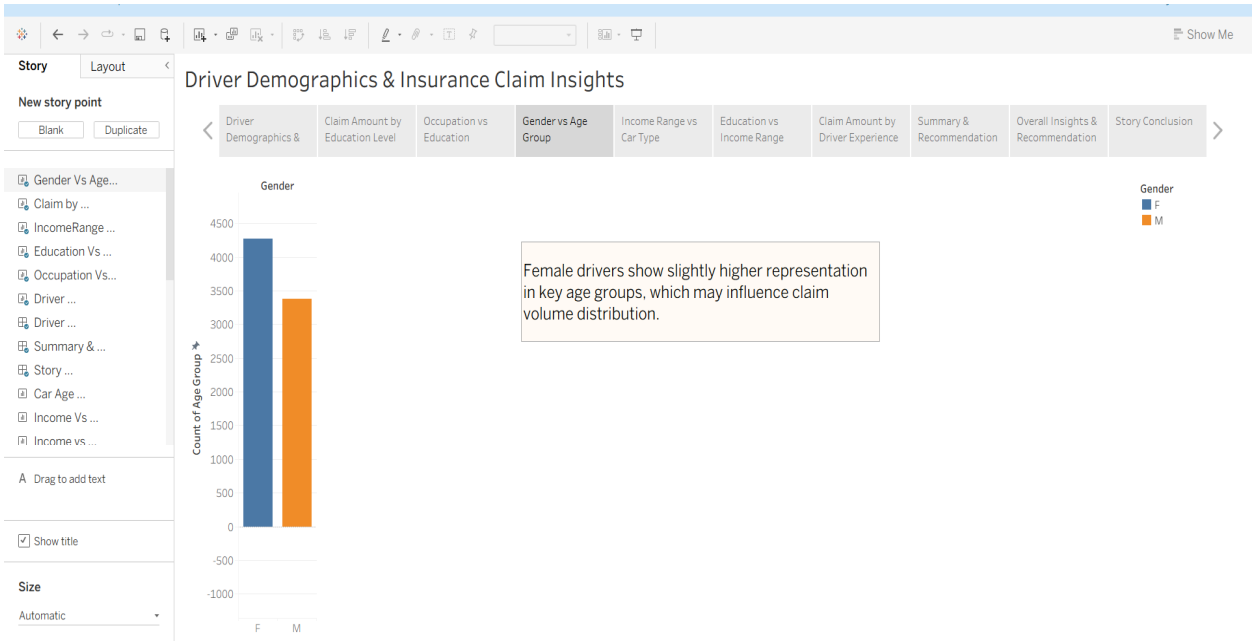
1.1 Claim by Education



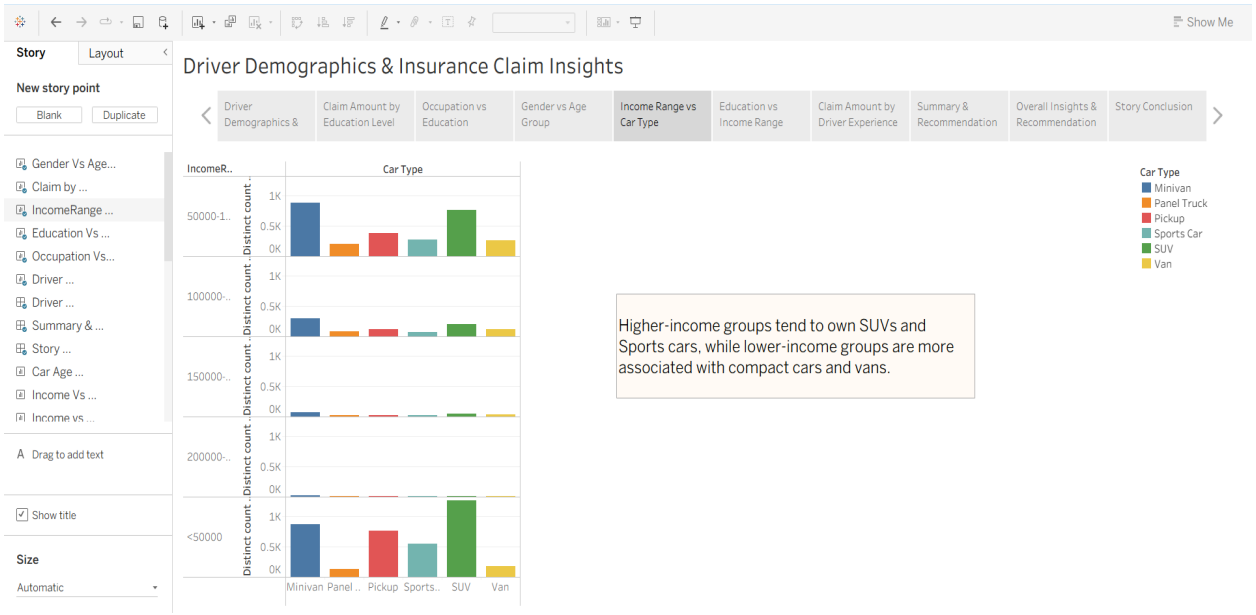
1.2 Occupation Vs Education



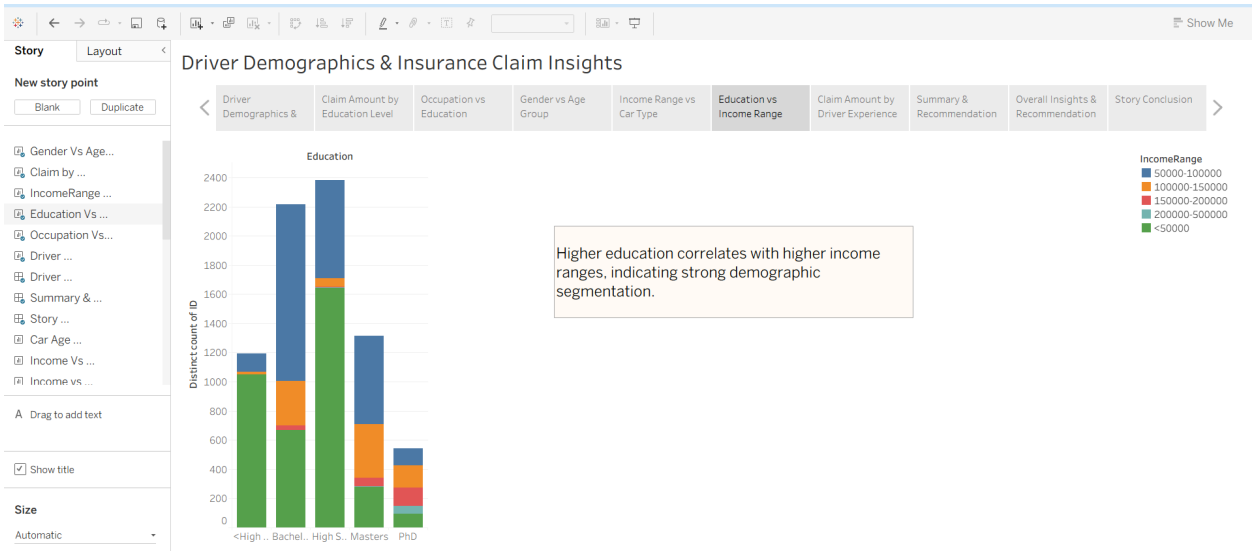
1.3 Gender Vs Age Group



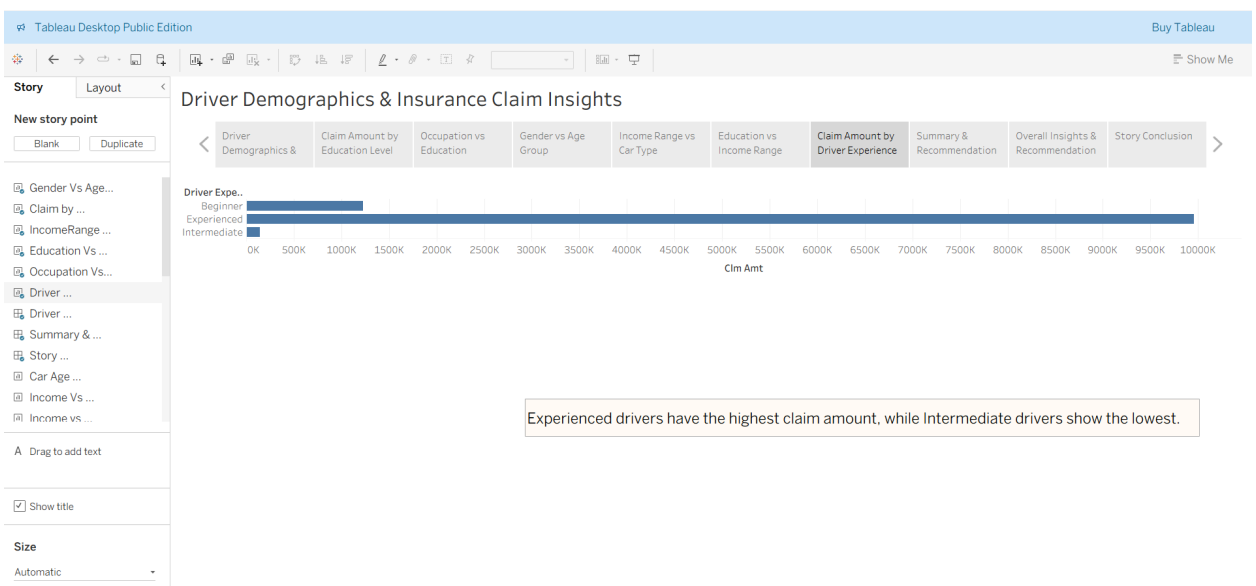
1.4 Income Range Vs Car Type



1.5 Education Vs Income Range



1.6. Claim Amount Vs Driver Experience Level



Summary & Recommendations

1.1 Claim by Education Level

Summary:
Drivers with High School and Bachelor's education levels contribute the highest overall claim amounts.
Recommendation:
Strengthen risk scoring and premium adjustments for these education segments to reduce claim exposure.

1.2 Occupation vs Education

Summary:
Blue Collar, Clerical, and Home Maker occupations are more common among lower-education groups, while Professionals and Managers dominate higher education levels.
Recommendation:
Design occupation-based premium brackets to more accurately reflect claim probability
.

1. 3 Gender vs Age Group

Summary:
Female drivers slightly outnumber males in the key age categories, influencing claim distribution patterns.
Recommendation:
Create targeted customer engagement or safety programs based on gender-age segments to improve claim outcomes.

1.4 Income Range vs Car Type

Summary:
Higher-income customers (200k+) generally own SUVs, Sports Cars, and Pickups, which are costlier in claim payouts.

Recommendation:
Implement vehicle-value-based premium adjustments to account for higher repair and replacement costs.

1.5 Education vs Income Range

Summary:
Higher education levels correlate strongly with higher income ranges, showing clear segmentation across the customer base.
Recommendation:
Use combined education–income segmentation for better targeting of premium plans and cross-sell opportunities.

1.6 Claim Amount by Driver Experience Level

Summary:
Experienced drivers have the highest claim amount compared to beginners and inexperienced drivers.
Recommendation
Experienced drivers are claiming more and adjusting insurance policies.

Overall Insights & Recommendations

Key Insights:
• High School and Bachelor's groups generate the highest claim amounts.
• Education strongly correlates with occupation and income level.
• Higher-income customers prefer high-value cars, impacting claim cost.
• Female drivers slightly outnumber male drivers across key age groups.

Business Recommendations:
• Strengthen risk assessment for High School & Bachelor groups.
• Offer tailored premium plans based on occupation and income segment.
• Introduce car-type based premium adjustments to manage claim costs.
• Develop demographic-specific marketing & safety programs.

Story Conclusion

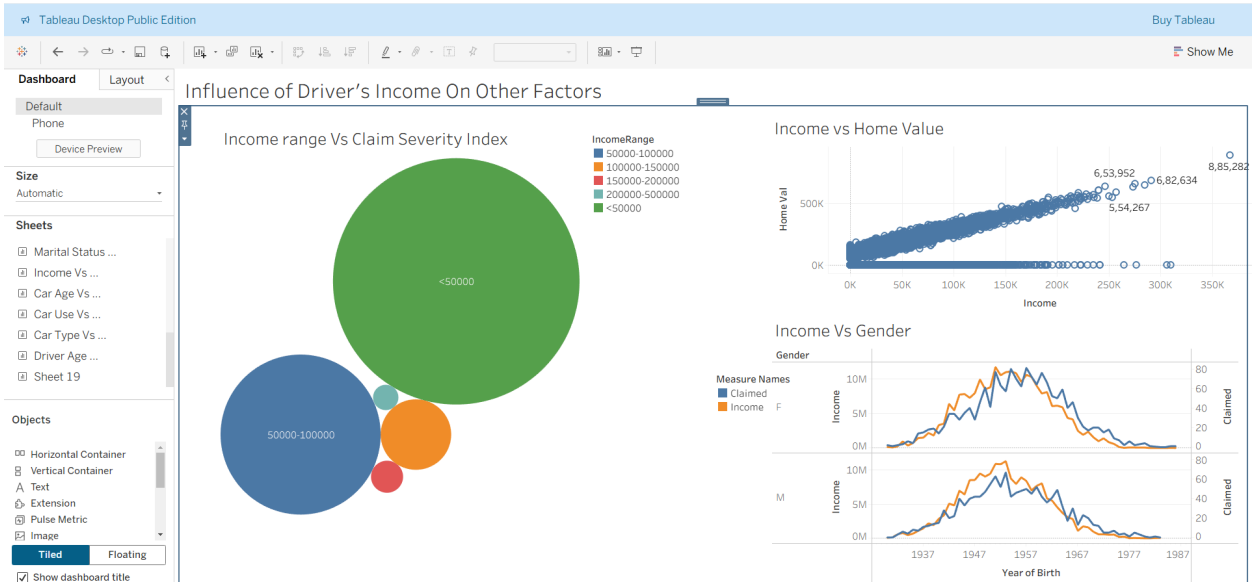
1. Education level, occupation, income, and car type collectively shape customer risk and claim behavior, showing clear patterns that insurers can use for better segmentation.

2. High School and Bachelor-educated drivers represent the largest share of claim amounts, indicating the need for more tailored pricing and risk management for these segments.

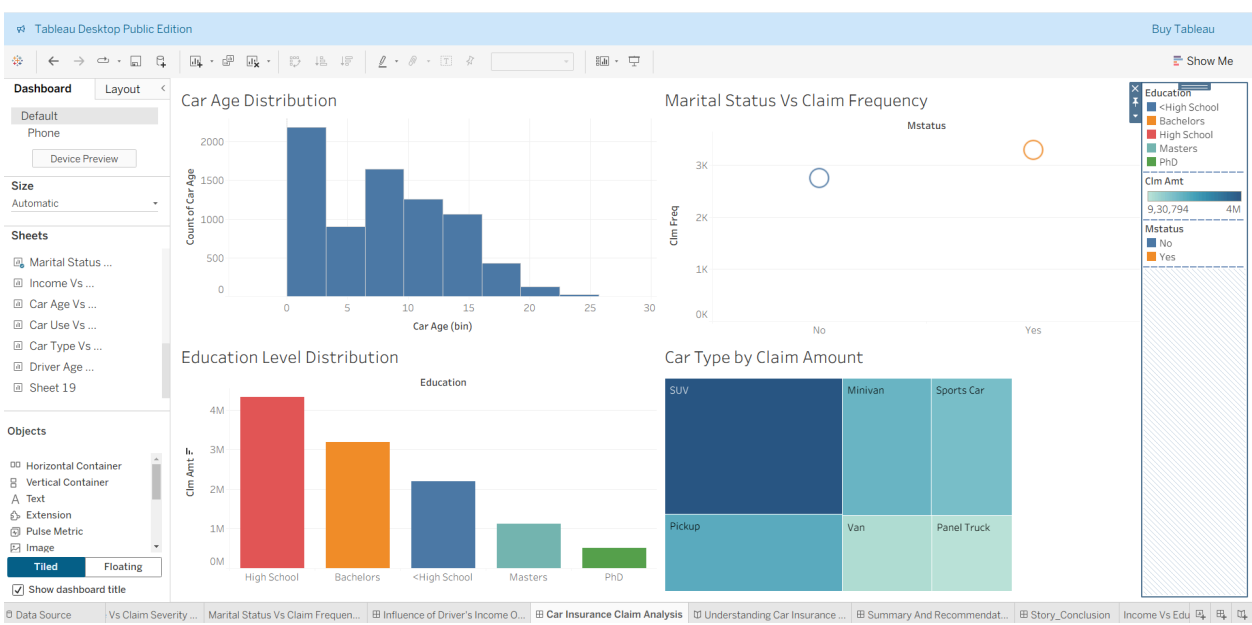
3. Higher-income customers tend to own high-value vehicles, which increases claim costs and highlights the importance of vehicle-value-based premium models.

4. Demographic factors such as gender and age group influence claim frequency, helping insurers focus on the right customer groups for safety initiatives and pricing refinement.

Creation of Influence of Driver's Income On Other Factors Dashboard

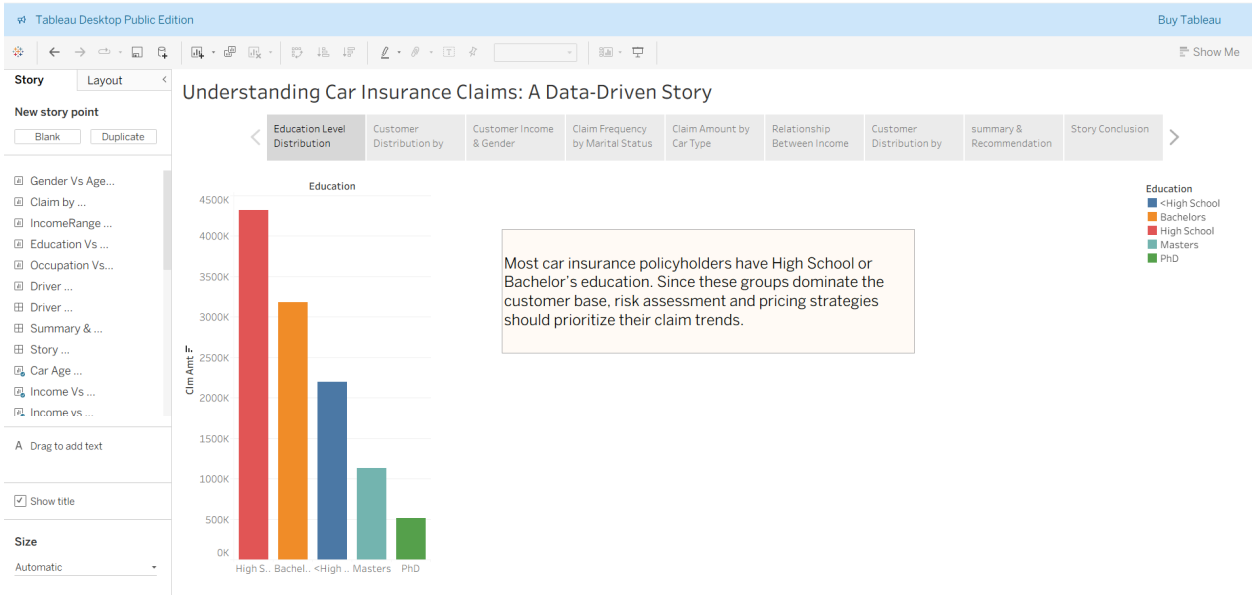


Creation of Car Insurance Claim Analysis DashBoard

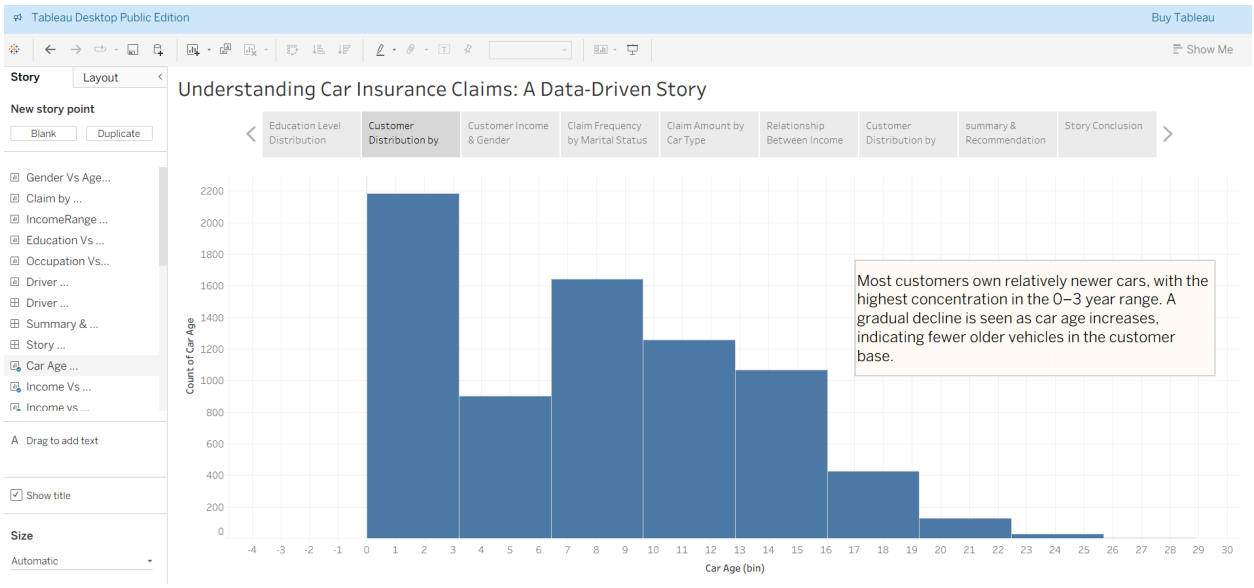


Creation Of Car Insurance Claims & Driver's Income StoryBoard

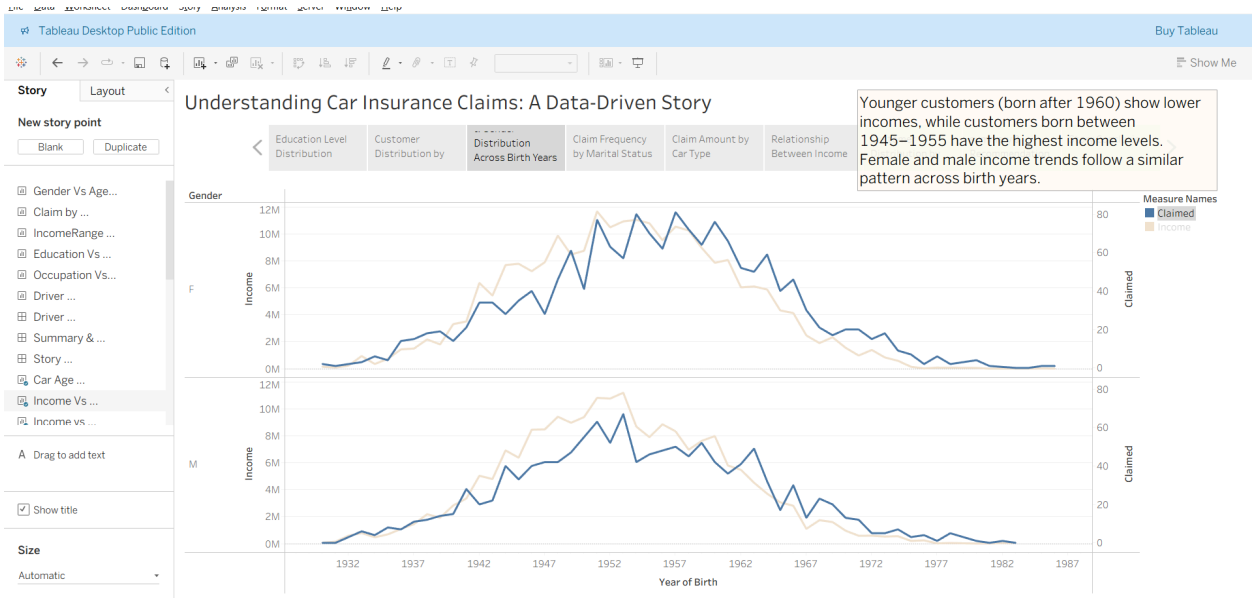
2.1 Education Level Distribution



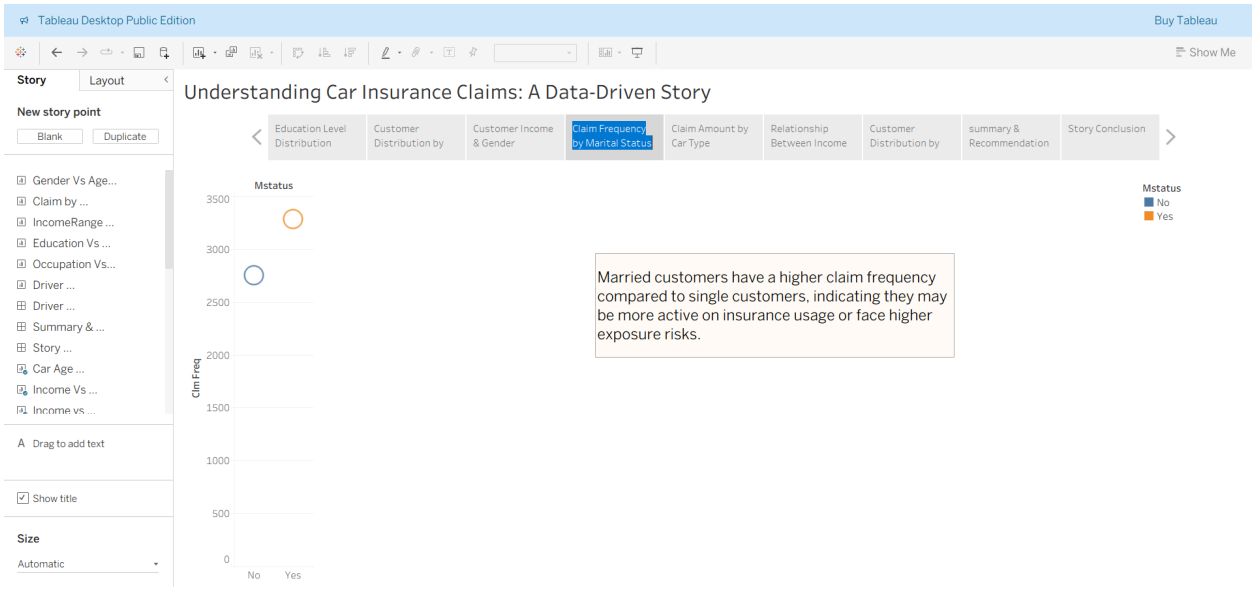
2.2 Customer Distribution by Car Age



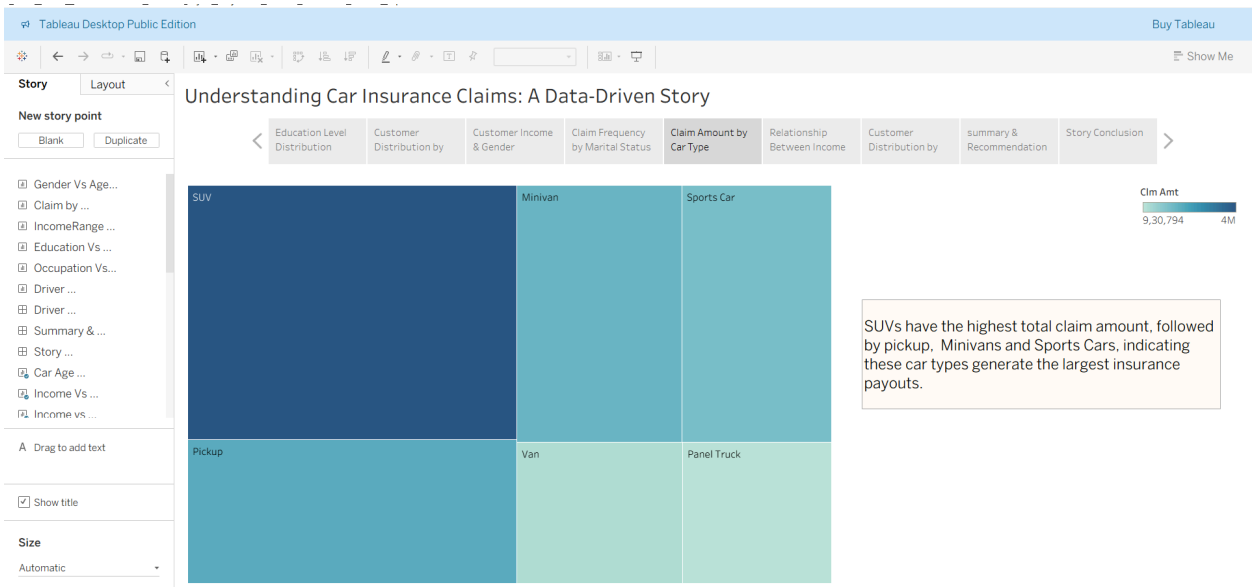
2.3 Customer Income & Gender Distribution Across Birth Years



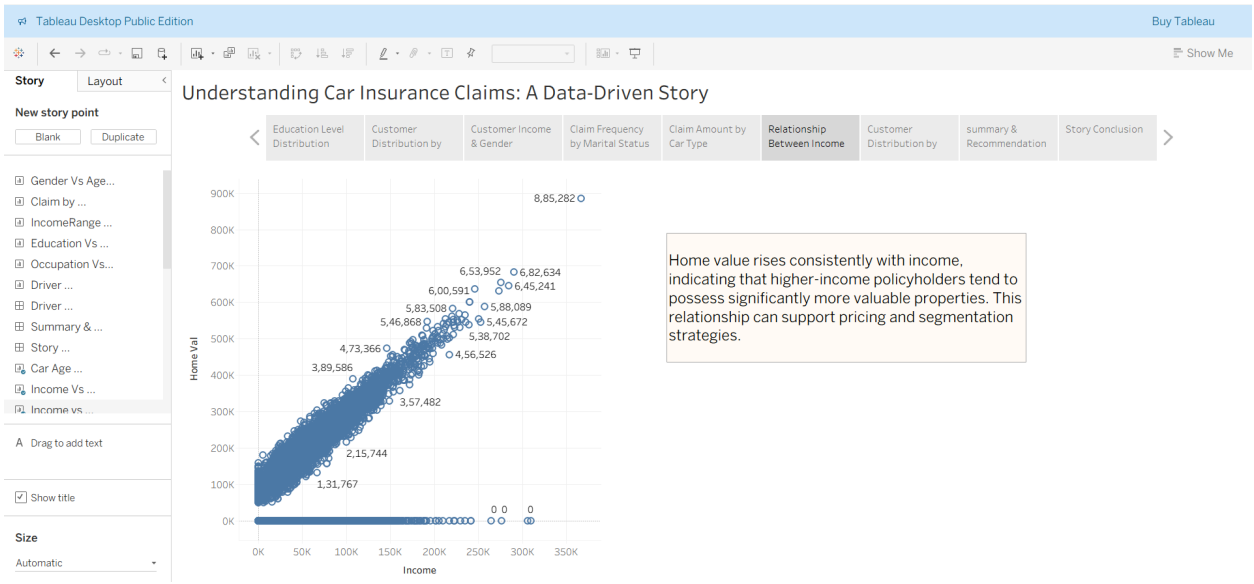
2.4 Claim Frequency by Marital Status



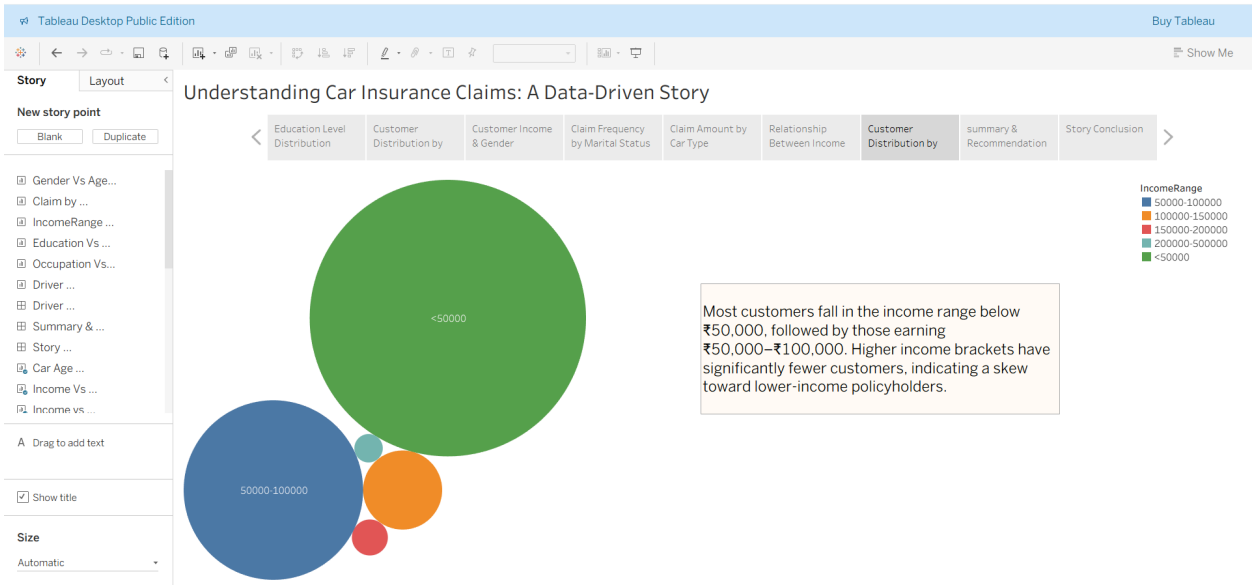
2.5 Claim Amount by Car Type



2.6 Relationship Between Income and Home Value



2.7 Customer Distribution by Income Range



Summary & Recommendations

2.1 Education Level Distribution

Summary:

Most policyholders have High School or Bachelor's education, making up the majority of the customer base.

Recommendation:

Target these education groups with tailored insurance offerings since they represent the highest customer volume.

2.2 Income Range vs Claim Severity Index

Summary:

Customers earning below ₹50,000 show the highest claim severity, indicating higher financial risk.

Recommendation:

Introduce risk-adjusted premium pricing or preventive programs for low-income customers.

2.3 Marital Status vs Claim Frequency

Summary:
Married customers have a slightly higher claim frequency than unmarried customers.

Recommendation:
Offer safe-driver discounts or bundled family policies to reduce claim frequency among married customers.

2.4 Income vs Gender

Summary:
Income distribution is similar for both genders, with no major income gap across the customer base.

Recommendation:
Design gender-neutral pricing and marketing campaigns since earning patterns are comparable.

2.5 Income vs Home Value

Summary:
Higher income customers own significantly higher value homes, showing strong positive correlation.

Recommendation:
Promote premium insurance products (car + home bundles) to high-income customers.

2.6 Car Type by Claim Amount

Summary:
SUVs contribute the highest claim amounts compared to other car types.

Recommendation:
Implement stricter risk assessment or premium adjustments for SUV owners.

2.7 Car Age Distribution

Summary:
Most customers own cars between 0–10 years, with very few older vehicles.

Recommendation:
Introduce special plans for older cars (10+ years) to tap into the small but underserved segment.

Story Conclusion

1. The customer base is dominated by middle-aged, mid-income individuals with moderate driving experience, and their behavior strongly influences claim outcomes.

2. High claim severity is concentrated among low-income groups and SUV owners, indicating clear pockets of financial and operational risk that require targeted pricing strategies.

3. Marital status and education show meaningful differences in claim frequency and severity, suggesting that demographic attributes can help refine customer segmentation and improve risk prediction.

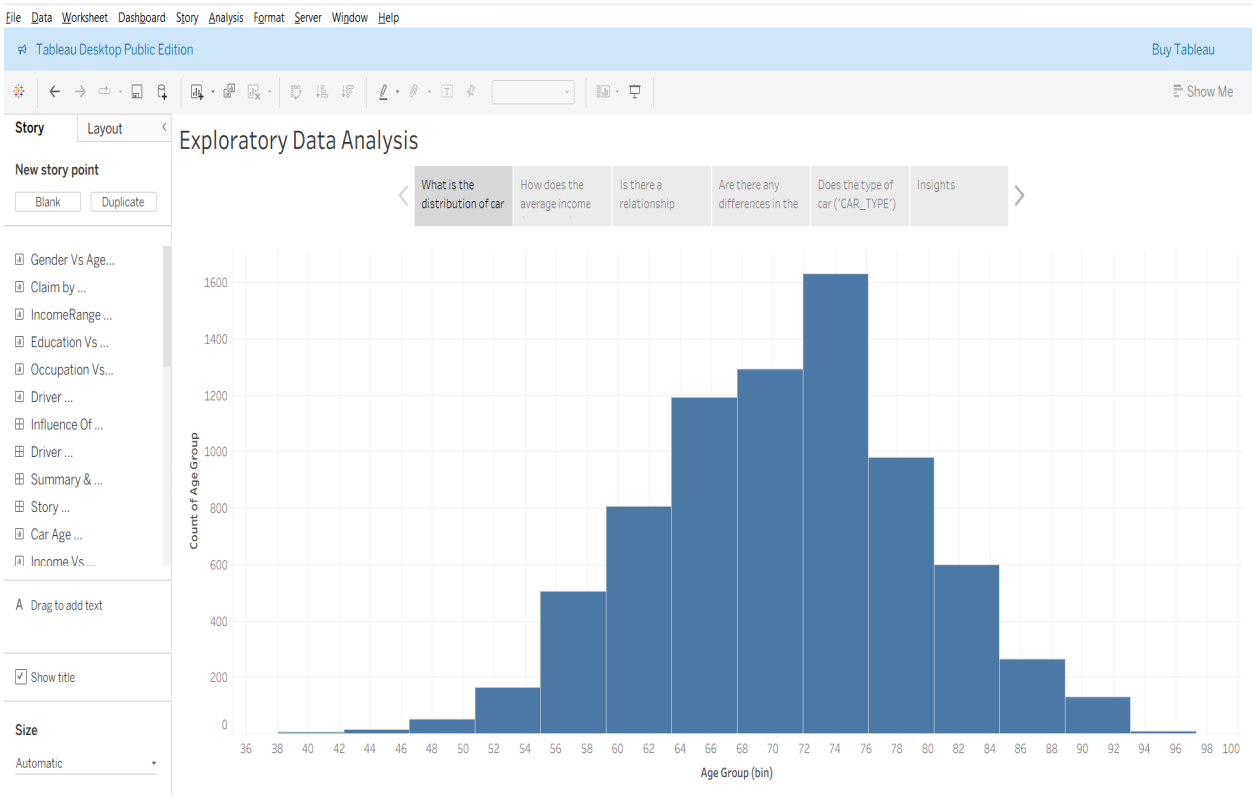
4. Vehicle-related factors—such as car age and type—play a major role in claim costs, emphasizing the need for risk-based premiums and preventive customer interventions.

5. Overall, the insights highlight where the insurer can optimize premium structures, reduce risk exposure, and design tailored offerings to improve profitability while ensuring fair and data-driven pricing.

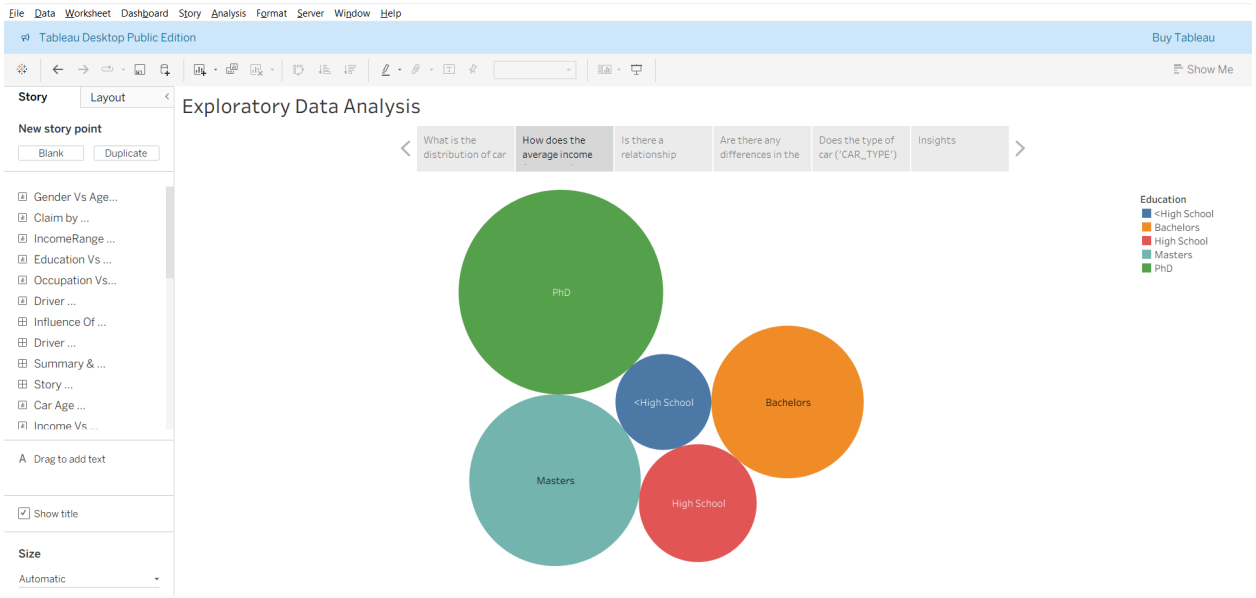
Exploratory Data Analysis

Creation Of EDA Charts DashBoard

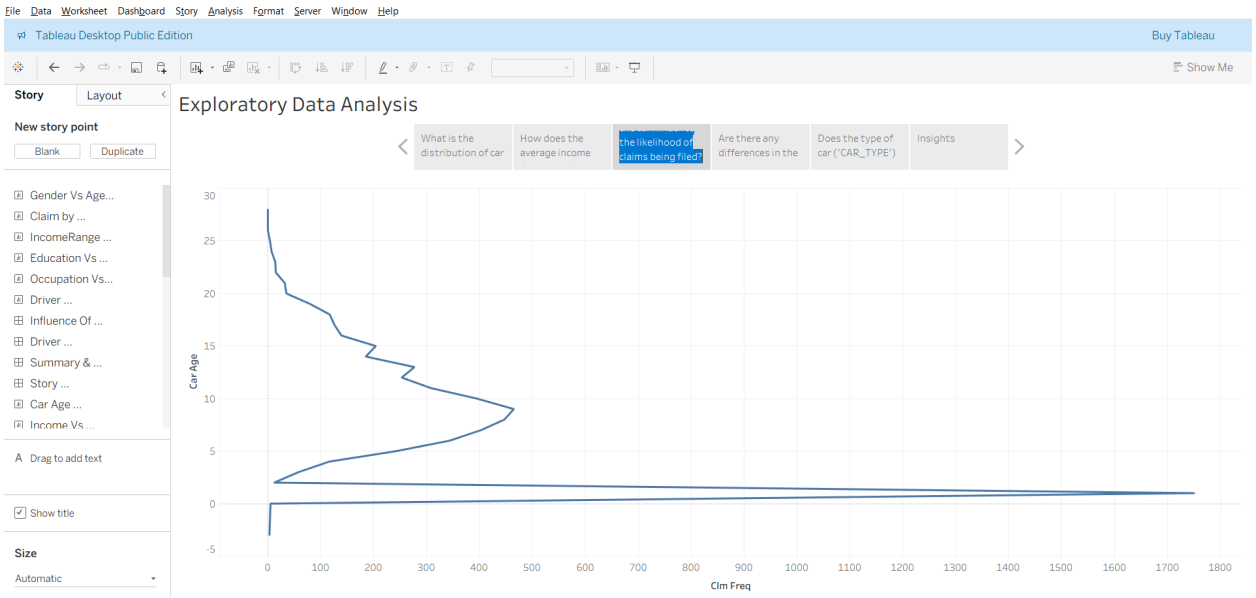
1. What is the distribution of car owners' ages (using the 'BIRTH' variable)? Are there any outliers or notable trends in the age distribution?



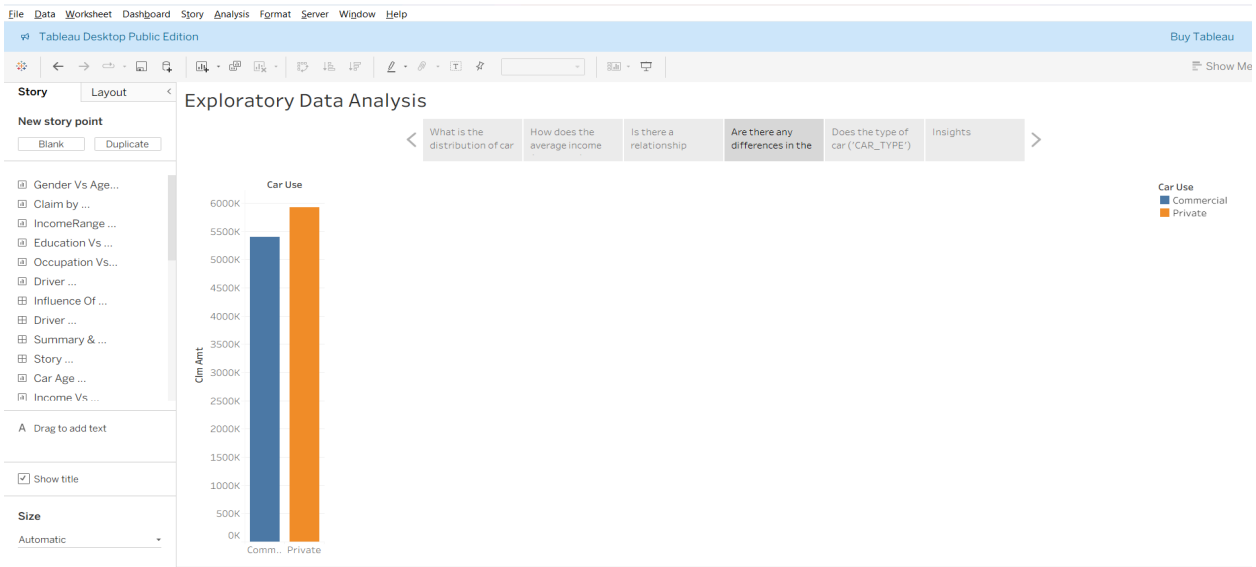
2. How does the average income ('INCOME') of drivers vary based on their education level ('EDUCATION')? Are there any specific education levels that tend to correlate with higher incomes?



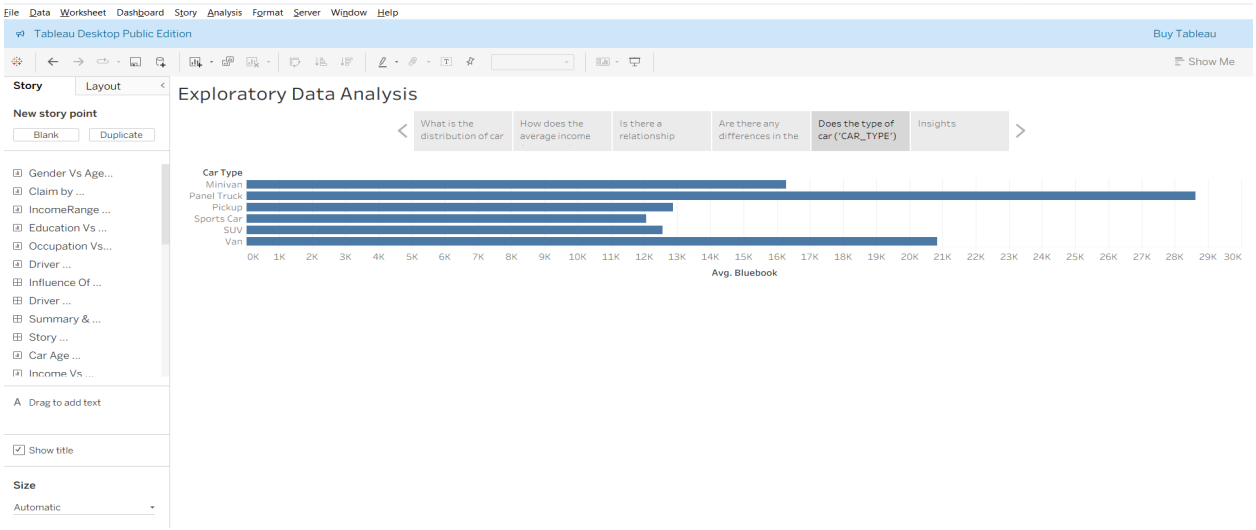
3. Is there a relationship between the age of the car ('CAR_AGE') and the frequency of claims ('CLM_FREQ')? Does the age of the car influence the likelihood of claims being filed?



4. Are there any differences in the distribution of claimed amounts ('CLM_AMT') between drivers who use their cars for different purposes ('CAR_USE')? How does this difference compare between gender ('GENDER') groups as well?



5. Does the type of car ('CAR_TYPE') have an impact on the average worth of the car ('BLUEBOOK')? Additionally, can you identify any patterns in terms of marital status ('MSTATUS') and car type ('CAR_TYPE') combinations?



Insights:

1. Driver Age Group Distribution
* The majority of car owners fall in the 65–75 age group.
* Very few drivers are in the younger (40–55) age range.
* A small number of drivers appear above 85+, which may be outliers.

2. Income vs Education
* Higher education levels earn higher income — PhD holders have the highest average income.
* Masters and Bachelors also show strong incomes, forming the middle-income segment.
* High School and Below High School earn the lowest, indicating a positive link between education and income.

3. Car Age vs Claim Frequency
* Newer cars (0–5 years) show the highest claim frequency.
* Claim frequency decreases as the car gets older.
* Very old cars (15+ years) show very low claim frequency, possibly due to reduced usage.

4. Car Use vs Claim Amount
* Private car owners have a higher total claim amount than commercial car owners.
* This suggests private vehicles file more or higher-value claims.
* Commercial vehicles show lower claim amounts, possibly due to stricter usage control or maintenance.

5. Car Type vs Bluebook Value
* Minivans have the highest Bluebook value among car types.
* Sports Cars and Pickups show mid-range values.
* Panel Trucks have the lowest value, indicating older or lower-cost vehicles.

Tableau Public URL

https://public.tableau.com/app/profile/chaitra.m4055/viz/Car_Insurance_Claim_17655247687020/InsightsforEachChart